



IOM205 Advanced Database Management

Individual Coursework Report

UniBooks Database

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ID2.2: The operation of the system

Step1: Log in

Upon double-clicking to open the Access database, a login page automatically appears, requiring users to enter their username and password before clicking the 'Login' button.



Figure 1: The login interface.

If entered incorrectly, the system will prompt an error message indicating an incorrect username and password. Users can re-enter the information or click the 'Exit' button on the right to exit the login interface.

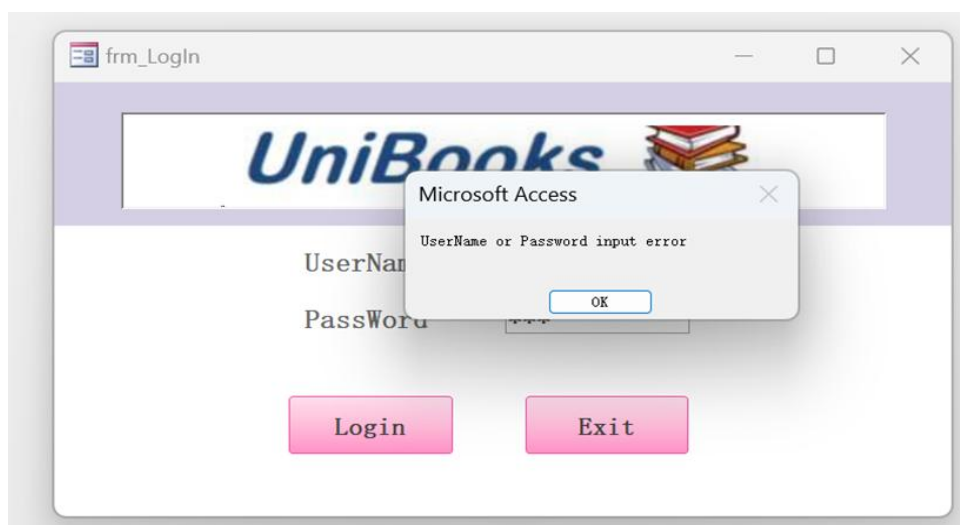


Figure 2: The login interface.

Step2: Step Into The Homepage

Upon successful login, users access the homepage in a pop-up window. They can exit using the 'X' or proceed to the next page by clicking the red button.

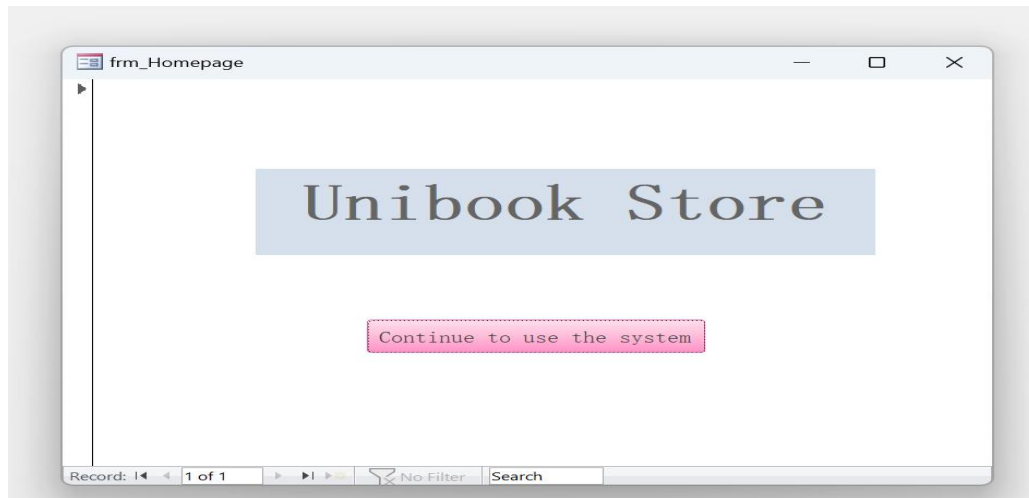


Figure 3: The homepage.

Step3: Come into the Menu

The interface encompasses most of the functionalities of the Unibook Store and is divided into three sections: All Orders Management, All Object Management, and Charts. The leftmost column displays detailed procurement and sales orders, the middle column manages personnel, and the Chart column visualizes business data. Clicking the red button provides detailed information on the details page.



Figure 4: The menu interface.

In addition, users can also perform automated searches through the left-hand menu page, selecting the desired query object and opening it with a double click.

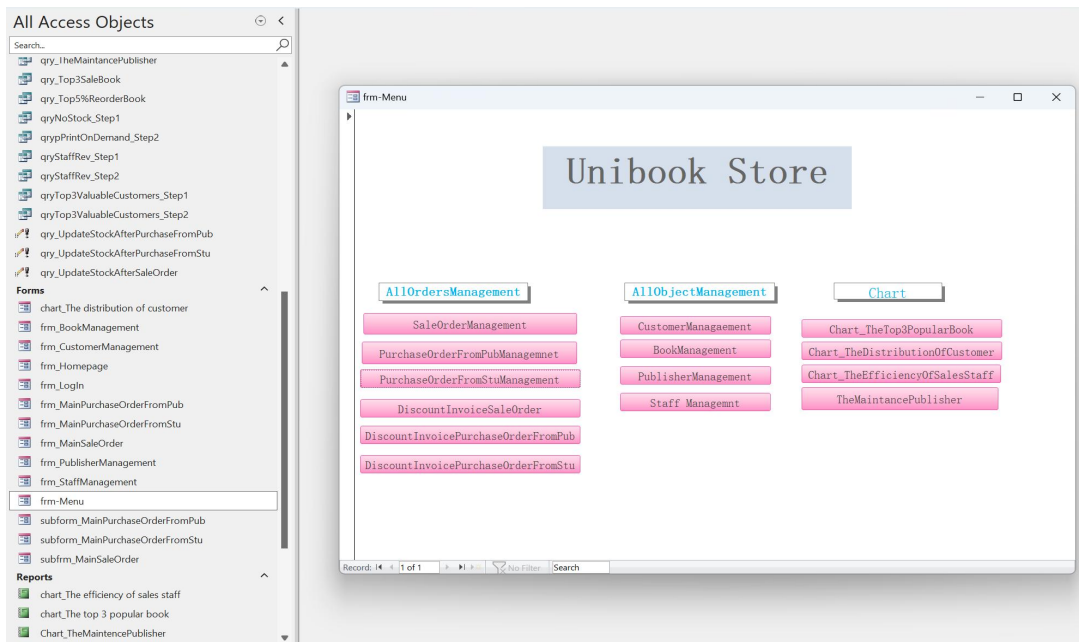


Figure 5: The menu interface.

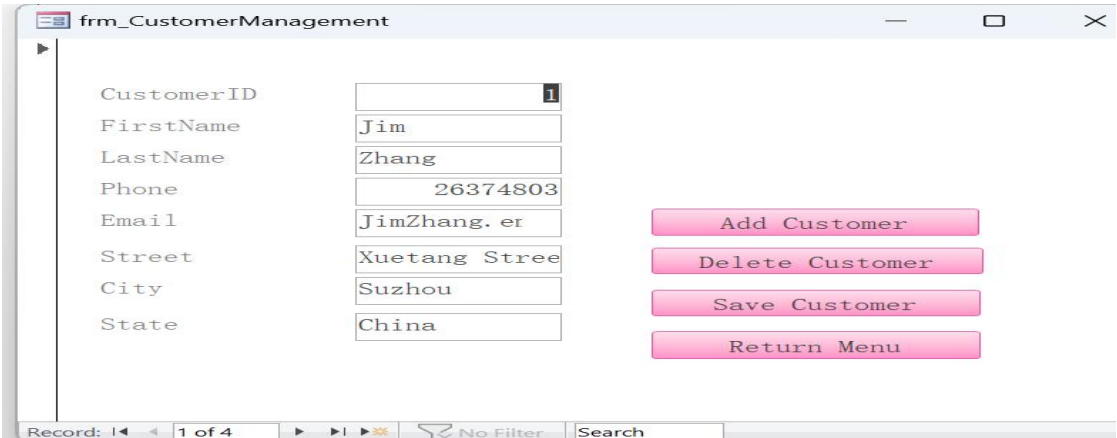
Step4: Add, delete and check operation

Choosing 'PurchaseOrderFromPubManagemnet' provides access to an interface with four red buttons for adding supplier orders, updating book quantities and order status, generating discounted invoices, and returning to the menu, with similar functionalities on other pages.

PurBookOrderFromPubID	PurchaseOrderFromPubID	ReorderQuantity	Book
1	1	2	Marketing
*(New)	1	0	

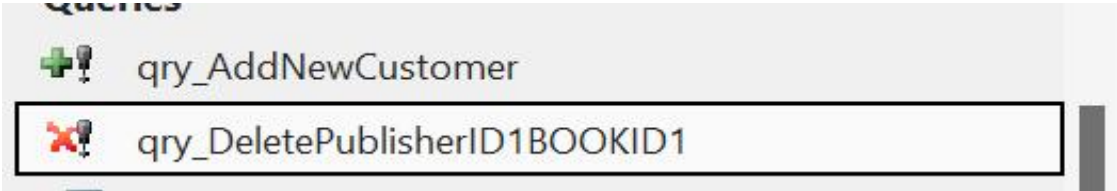
Figure 6: The main purchase order from publisher form.

The system also enables deletion via form and query.



The screenshot shows a Windows-style application window titled "frm_CustomerManagement". It contains a form with the following fields and values: CustomerID (empty), FirstName (Jim), LastName (Zhang), Phone (26374803), Email (JimZhang.er), Street (Xuetang Stree), City (Suzhou), and State (China). To the right of the form are four pink buttons: "Add Customer", "Delete Customer", "Save Customer", and "Return Menu". At the bottom of the window, there is a status bar showing "Record: 1 of 4", a "No Filter" indicator, and a "Search" button.

Figure 7: The customer management form.



The screenshot shows a list of queries. The first query is "qry_AddNewCustomer" with a green plus icon. The second query is "qry_DeletePublisherID1BOOKID1" with a red X icon and is highlighted with a black border.

Figure 8: The add and delete query.

ID2.3: Two function

2.3.1 Inventory Management Optimization

To augment inventory management efficiency, the system has incorporated new fields into the tblBook: Stock Level, Reorder Level, and Target Level. These additions enable the system to prompt purchasing actions when inventory levels reach the restocking threshold and determine order quantities by comparing the differential between Target and Stock Level, thereby maintaining optimal inventory levels (Chiru, 2015).

Through the integration between tblBook and tblPublisher and their connection with tblPurchaseOrderFromPublisher, employees can swiftly locate publisher information via BookID for procurement, streamlining the response process to demands (Jiang & Naumann, 2020).

The newly integrated ProcessStatus field into the tblSaleOrder, tblPurchaseOrderFromPublisher, and tblPurchaseOrderFromStudent tables enhances

the tracking of order status, reducing the time lapse from procurement to sales and elevating response rates and customer satisfaction.

These modifications allow the system to forecast customer needs more accurately, circumvent inventory issues, ensure the fluidity of capital, and fortify brand loyalty. These intelligent enhancements bolster the efficiency of UniBooks' order processing, contributing significantly to the business's growth trajectory.

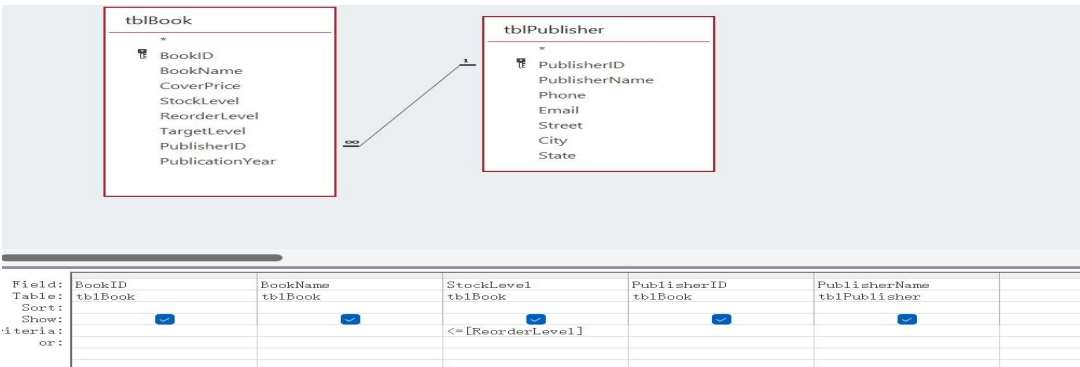


Figure 9: The purchase book order from publisher.

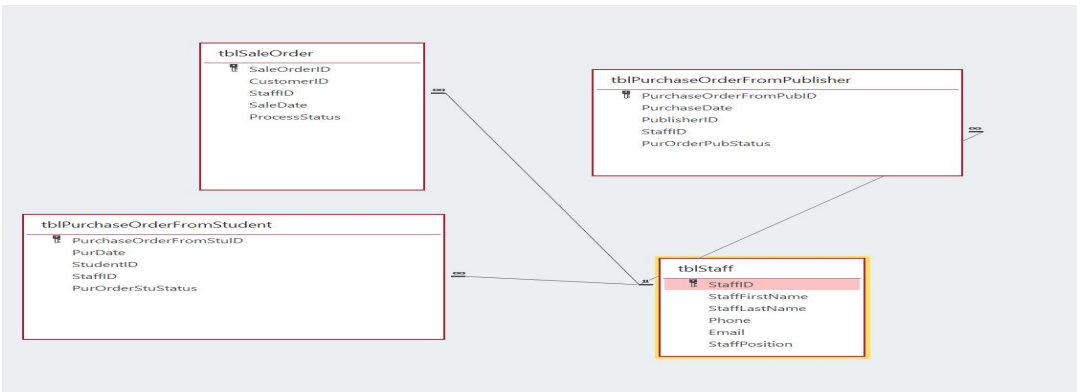


Figure 10: The ER model for all orders.

2.3.2 On-Demand Management Reports

System's management reporting feature is committed to capturing and analyzing detailed sales and procurement data, ensuring that order activities are recorded comprehensively and promptly at each critical juncture. The system meticulously categorizes orders into sales, supplier, and student purchase orders, producing time-series analytical reports encompassing key business metrics.

The clarity in the discount settings also plays a crucial role in facilitating sales

revenue and procurement. Discounts for sales and purchase orders to suppliers are determined based on the order size, with additional discounts of 50% and 10%, respectively. For student orders, a uniform discount of 20% is applied.

The reports incorporate a complex discount querying mechanism and dynamic visualization charts, significantly enhancing management's insight into business trends, profit, and cost metrics, core customer purchasing behavior, and employee performance. This comprehensive understanding aids in formulating future procurement and sales strategies and maintaining effective relationships.

PurBookOrderFromPubID	PurchaseOrderFromPubID	ReorderQuantity	Book
1	1	1	Marketing

Figure 11: The purchase order from publisher form.

BookName	CoverPrice	ReorderQuantity	Cost
Marketing	\$15.00	2	¥30.00

Discount: 0
Total: 3

Figure 12: The discount invoice for the purchase order from the publisher.

ID2.4 Two Non-Trivial queries

2.4.1 Top 3 Valuable Customer

The system's multi-stage queries encompass advanced querying techniques and profound business insights.

The first-stage query captures the total sales revenue per customer, providing fundamental data for identifying key customers, which is crucial for resource optimization and relationship maintenance.

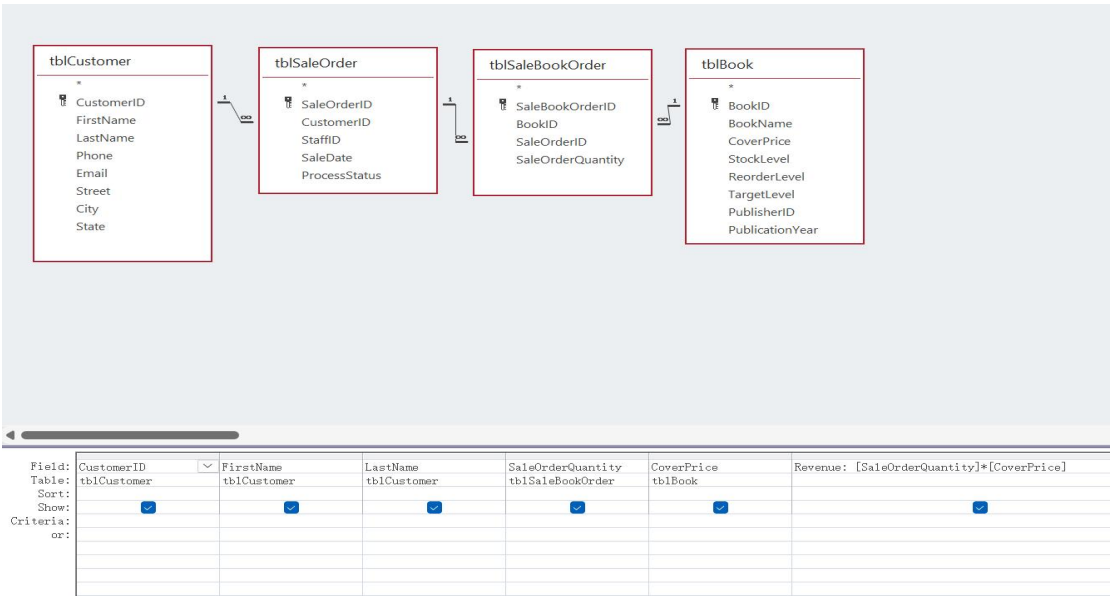


Figure13: The step 1 of top 3 valuable customer query design.



Figure 14: The output of figure 13.

The second query identifies the top three revenue-generating customers, assisting the organization in concentrating efforts on the most valuable clientele, thereby enhancing satisfaction and loyalty. These two queries have a direct impact on marketing strategy

and resource allocation, constituting a critical aspect in the implementation of targeted customer relationship management and growth strategies.

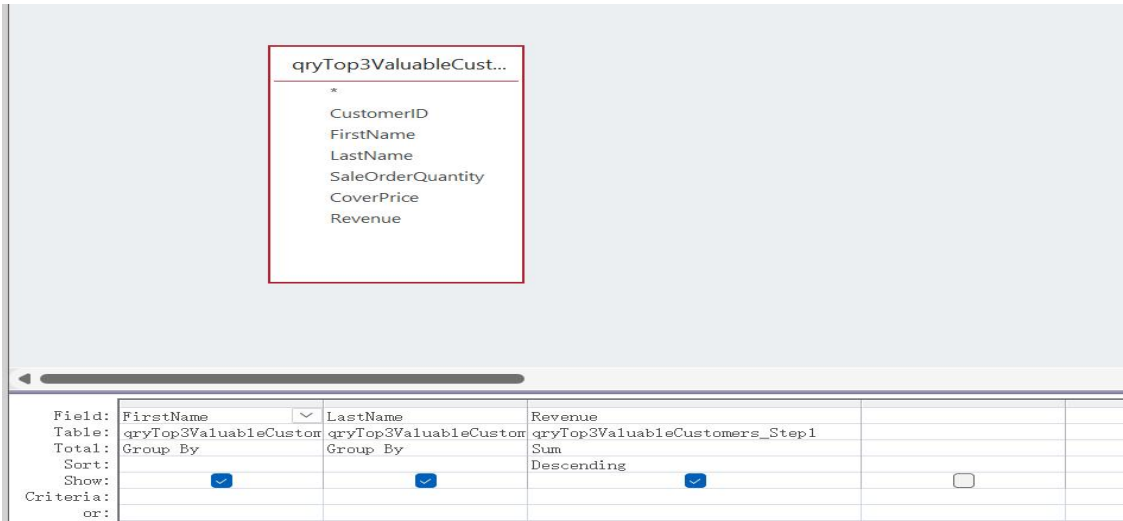


Figure 15: The step2 of top 3 valuable customer query design.

FirstName	LastName	RevenueOfSum
Mike	Black	¥455.00
Jim	Zhang	¥280.00
Lily	Lee	¥233.00

Figure 16: The output of the figure15.

2.4.2 The sale ability of staff

SQL code step 1:

```
SELECT tblSaleOrder.StaffID, tblStaff.StaffFirstName, tblStaff.StaffLastName,
tblSaleBookOrder.BookID,tblSaleBookOrder.SaleOrderQuantity,tblBook.CoverPrice,
[SaleOrderQuantity]*[CoverPrice] AS Sales
FROM tblStaff INNER JOIN (tblSaleOrder INNER JOIN (tblBook INNER JOIN
tblSaleBookOrder ON tblBook.BookID = tblSaleBookOrder.BookID) ON
tblSaleOrder.SaleOrderID = tblSaleBookOrder.SaleOrderID) ON tblStaff.StaffID =
tblSaleOrder.StaffID;
```

This SQL query retrieves data from multiple tables, including the employee (tblStaff),

sales order (tblSaleOrder), book (tblBook), and sales book order (tblSaleBookOrder) tables. It employs INNER JOIN statements to establish associations between these tables. The query selects employee ID, employee first name, employee last name, book ID, sales order quantity, book cover price, and the computed total sales amount (Sales). Ultimately, the query returns sales data, comprising information about the selling employees and related sales orders and books data.

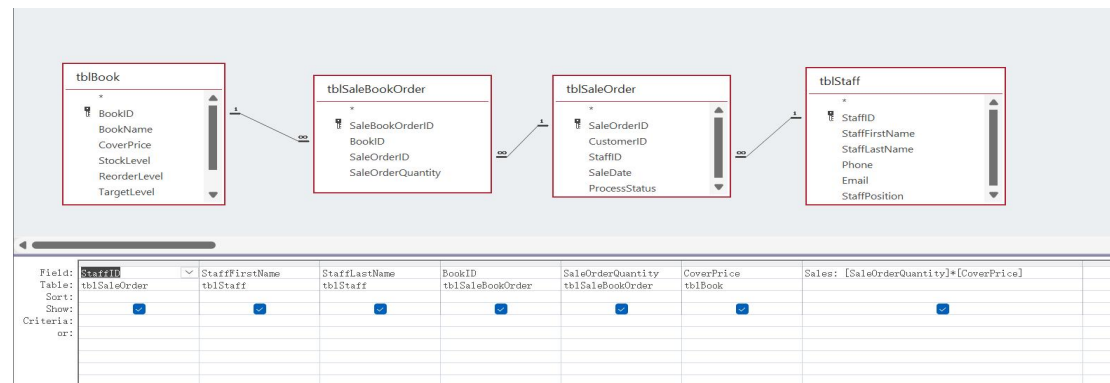


Figure 17: The step 1 of the sailing ability of staff.

StaffID	StaffFirstName	StaffLastName	BookID	SalesOrderQuantity	CoverPrice	Sales
4	Joy	Green	1	2	\$15.00	¥30.00
4	Joy	Green	3	3	\$20.00	¥60.00
5	Jimi	Tom	2	7	\$15.00	¥105.00
4	Joy	Green	5	10	\$25.00	¥250.00
5	Jimi	Tom	7	4	\$32.00	¥128.00
5	Jimi	Tom	4	6	\$20.00	¥120.00
5	Jimi	Tom	3	10	\$20.00	¥200.00
5	Jimi	Tom	3	1	\$20.00	¥20.00
5	Jimi	Tom	2	5	\$15.00	¥75.00
4	Joy	Green	10	4	\$40.00	¥160.00

Figure 18: The output of figure 17.

SQL code Step2:

```
SELECT qryStaffRev_Step1.StaffFirstName, qryStaffRev_Step1.StaffLastName,
Sum(qryStaffRev_Step1.Sales) AS SalesOfSum
FROM qryStaffRev_Step1
GROUP BY qryStaffRev_Step1.StaffFirstName, qryStaffRev_Step1.StaffLastName
ORDER BY Sum(qryStaffRev_Step1.Sales) DESC;
```

This SQL query is based on the results of the first query (potentially named qryStaffRev_Step1). It further aggregates sales data for employees, grouping them by

their first and last names. Subsequently, it calculates the total sales amount for each employee, aliasing it as "SalesOfSum." Finally, the query results are ordered in descending order of sales total, listing the employees with the highest sales totals at the top. This query serves the purpose of identifying employees with the best sales performance.

This query facilitates the identification of top-performing employees, aiding in fair rewards, customized training, and resource optimization. It enhances overall sales efficiency and informs strategic decision-making for business success.

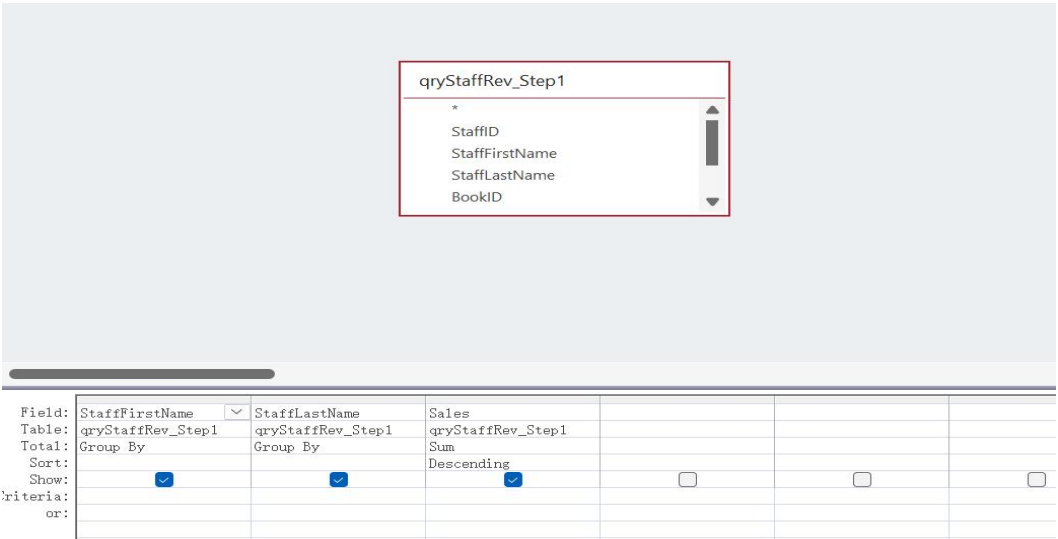


Figure 19: The step2 of the sailing ability of staff.

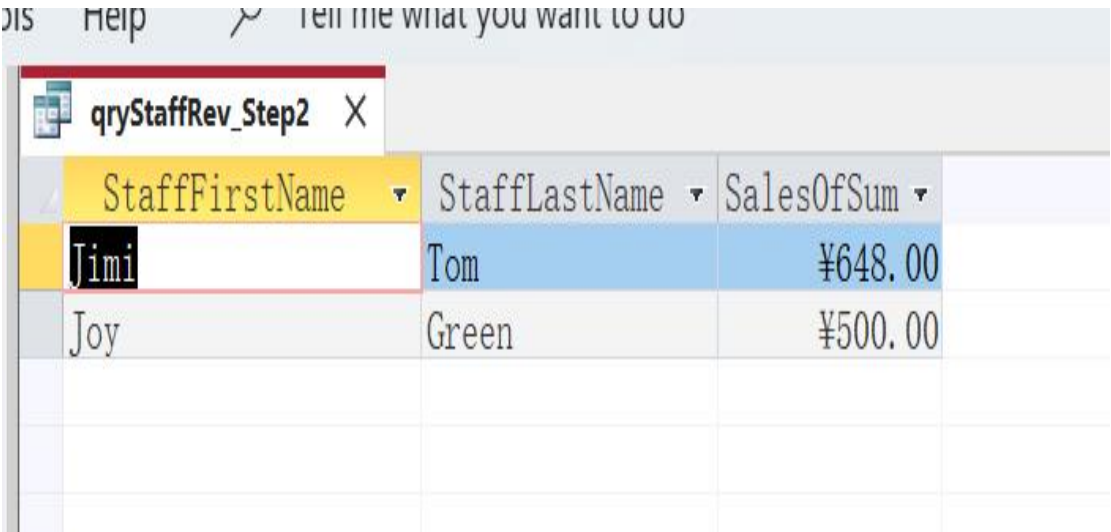


Figure 20: The output of figure 19.

ID2.5: Two management report

2.5.1 The discount Invoice of Sale order

This report is valuable for senior management as it provides detailed information about sales orders, including customer details, dates, quantities, and total sales revenue. Such data enables senior management to monitor sales activities, assess trends, and evaluate employee sales performance. Furthermore, the report's discount and total information can assist senior management in understanding profit margins and formulating or adjusting pricing strategies.

SaleBookOr	SaleOrderID	SaleOrderQuantity	Book
4	3	10	Digital Marke
(New)	3	0	

Figure 21: The main sale order form.

BookName	SaleOrderQuantity	CoverPrice	Cost
Digital Marke	10	\$25.00	¥250.00

Discount: 5
Total: 118.75

Figure 22: The discount invoice of sale order.

2.5.2 The discount invoice of Purchase order From Student

This report reflects the company's book procurement from students. It assists senior management in monitoring the cost of purchasing used books, evaluating the profitability of buyback programs, forecasting textbook demand, and providing insights into market trends. Additionally, it enhances relationships with students and promotes repeat student purchases.

frm_MainPurchaseOrderFromStu

PurchaseOrder3

StudentLili

StaffAndy

PurDate2023/6/26

Update and Save

Generate Invoice

Return Menu

PurchaseFromStuOrderLine

PurBookOrderFromStuID	PurchaseOrderFromStuID	PurOrderFromStuQuantity	Book
3	3	9	Economics
*(New)	3	0	

Figure 23: The main purchase order from student form.

PurchaseOrder3

PurDate2023/6/26

StuFirstNameLili

StuLastNameWhite

BookName	CoverPrice	PurOrderFromStuQuantity	Cost
Economics	\$20.00	9	¥180.00

DiscountRate0.2

Total36

Figure 24: The discount invoice of Figure 23.

ID2.6 One non-trivial chart-The maintenance publisher

This table and chart display the quantities of books ordered by the company from various publishers. User can swiftly identify which publishers provide the highest and lowest quantities of books through data aggregation and pie charts. The chart aids senior management in assessing and optimizing relationships with publishers. The charts make the data more intuitive and understandable, guiding procurement decisions, balancing inventory, and considering the importance of each publisher in future procurement plans. The chart is crucial for maintaining robust supply chain relationships, inventory diversity, and cost-effectiveness analysis.

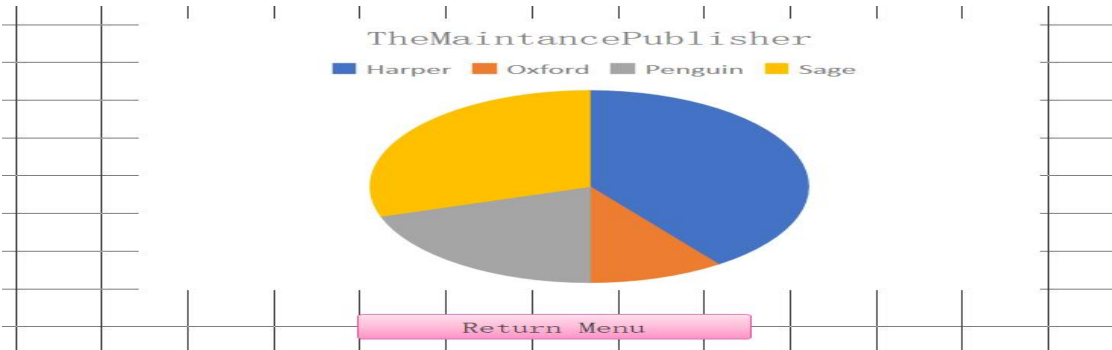


Figure 25: The one non-trivial chart-The maintenance publisher.

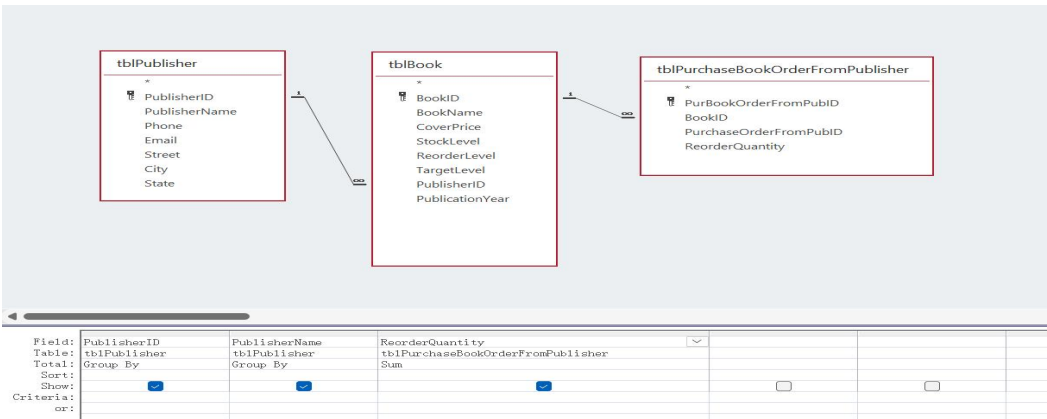


Figure 26: The query design of Figure 25.

PublisherID	PublisherName	ReorderQuantityOfSum
1	Oxford	12
2	Penguin	37
3	Sage	30
4	Harper	15

Figure27: The output of query of Figure26.

ID2.7 Testing

2.7.1 Select query

Query books with a Stock Level of 0.

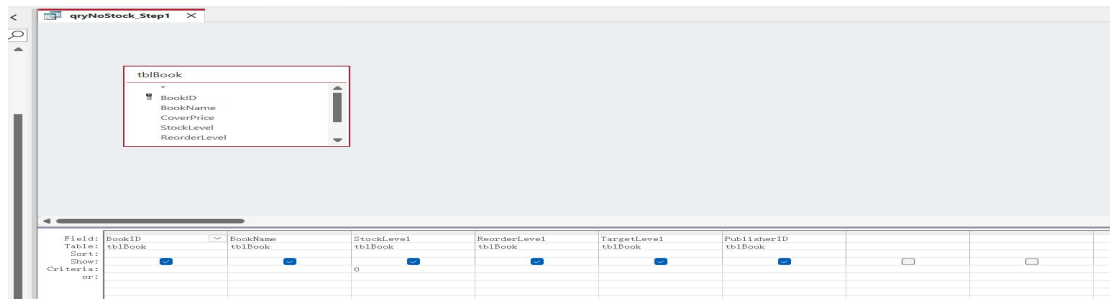


Figure28: The query of stock level=0.

The screenshot shows the data table for 'qryNoStock_Step1'. The table has columns: BookID, BookName, CoverPrice, StockLevel, ReorderLevel, TargetLevel, PublisherID, and Publication. The data is as follows:

BookID	BookName	CoverPrice	StockLevel	ReorderLevel	TargetLevel	PublisherID	Publication
1	Marketing	\$15.00	33	20	25	1	2010
2	Database	\$15.00	24	20	25	2	2011
3	Economics	\$20.00	24	20	25	2	2010
4	Accountancy	\$20.00	35	20	25	3	2014
5	Digital Marketing	\$25.00	25	20	25	3	2015
6	Big Data	\$25.00	20	20	25	4	2016
7	Management	\$32.00	32	20	25	5	2017
8	Ethics	\$35.00	19	20	25	5	2012
9	GDPR	\$40.00	20	20	25	4	2010
10	Maths	\$40.00	19	20	25	4	2009
11	Language	\$25.00	35	20	25	5	2009
12	Humanities	\$30.00	23	20	25	3	2008
13	AI	\$10.00	25	0	0	1	2011
14	Fishing	\$20.00	25	10	20	2	2009
(New)		\$0.00	0	0	0	0	0

Figure 29: The book table.

Actual and assumed outcomes are blank.

The screenshot shows the query editor for 'qryNoStock_Step1'. The query is defined as:
Field: BookID, Table: tblBook, Sort: StockLevel, Show: StockLevel, Criteria: 0.
The 'tblBook' table structure is shown in the top right: BookID, BookName, CoverPrice, StockLevel, ReorderLevel, TargetLevel, PublisherID.

Figure30: The actual output of query when stock level=0.

Query which of the books with a stock level of 0 are on-demand orders and which are reorders. Both the expected and actual results are blank.

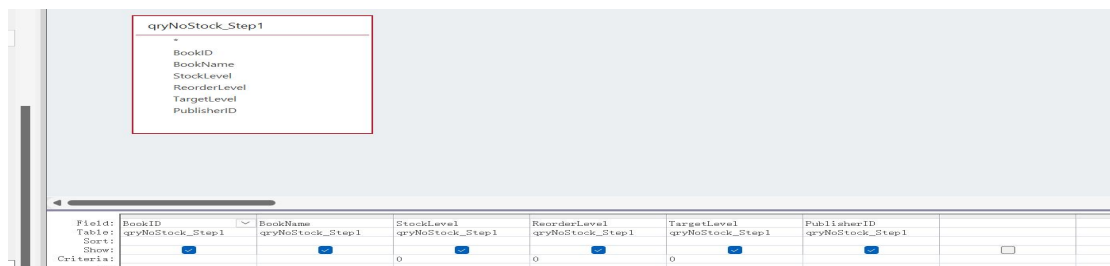


Figure 31: The all 0 condition query.

The screenshot shows the data table for 'qryNoStock_Step1'. The table has columns: BookID, BookName, StockLevel, ReorderLevel, TargetLevel, and PublisherID. The data is as follows:

BookID	BookName	StockLevel	ReorderLevel	TargetLevel	PublisherID
(New)		0	0	0	0

Figure 32: The actual output of Figure 31.

2.7.2 Form update and save

In the absence of added data, the existing data is as follows.

PurBookOrderFromPubID	BookID	PurchaseOrderFromPubID	ReorderQuantity	Click to Add
1	1	1	2	
14	2	15	17	
16	2	19	10	
17	3	20	10	
18	4	21	10	
19	9	22	10	
20	9	23	5	
21	11	24	10	
22	4	25	10	
23	1	26	10	
(New)	0	0	0	

Figure33: The purchase book order from publisher.

PurchaseOrderFromPubID	PurchaseDate	PublisherID	StaffID	PurOrderPubStatus	Click to Add
1	2023/12/8	1	1	✓	
15	2023/9/11	2	2	✓	
19	2023/9/21	2	2	✓	
20	2023/9/13	2	2	✓	
21	2023/12/4	3	2	✓	
22	2023/5/16	4	2	✓	
23	2023/9/13	4	2	✓	
24	2023/9/5	3	2	✓	
25	2023/11/27	3	2	✓	
26	2023/12/4	1	2	✓	
(New)		0	0		

Figure 34: The purchase order from publisher.

PurchaseOrderFrom

27

PurchaseDate

2023/12/5

Staff

Blue

Publisher

Xinhua

CreateAnOrderFromPub

Update and Save

Generate Invoice

Return Menu

PurchaseFromPubOrderLine

PurBookOrderFromPubID	PurchaseOrderFromPubID	ReorderQuantity	Book
24	27	10	Management
(New)	27	0	

Figure 35: The main purchase order from publisher form.

Now, within 'frm_Main PurchaseOrderFromPub,' a new record has been added, as shown in the figure. Consequently, in the 'tblPurchaseOrderFromPublisher' table, a new entry with 'PurBookPOrderFromPubID' of 27, 'PurchaseDate' of 2023/12/5, 'Publisher ID' of 5, 'StaffID' of 2, and 'PurOrderPubStatus' of 'No' has been created.

Simultaneously, within 'tblPurchaseOrderFromPublisher,' another entry has been

added with 'PurBookOrderFromPubID' of 24, 'BookID' of 7, 'PurchaseOrderFromPubID' of 27, and 'ReorderQuantity' of 10.

Actual and assumed outcomes are consistent.

PurchaseOrderFromPubID	PurchaseDate	PublisherID	StaffID	PurOrderPubStatus
15	2023/12/8	1	1	☒
19	2023/9/11	2	2	☒
20	2023/9/21	2	2	☒
21	2023/9/13	2	2	☒
22	2023/12/4	3	2	☒
23	2023/5/16	4	2	☒
24	2023/9/13	4	2	☒
25	2023/9/5	3	2	☒
26	2023/11/27	3	2	☒
27	2023/12/4	1	2	☒
(New)	2023/12/5	5	2	☐
(New)		0	0	☐

Figure 36: The purchase order from publisher after adding data.

PurBookOrderFromPubID	BookID	PurchaseOrderFromPubID	ReorderQuantity
1	1	1	2
14	2	15	17
16	2	19	10
17	3	20	10
18	4	21	10
19	9	22	10
20	9	23	5
21	11	24	10
22	4	25	10
23	1	26	10
24	7	27	10
(New)	0	0	0

Figure 37: The purchase book order from publisher after adding data.

Prior to the update, the inventory for the book 'Management' is 22. After the update, it will be 32, and the order status will also be Yes.

BookID	BookName	CoverPrice	StockLevel	ReorderLev	TargetLeve	PublisherID	Publication
1	Marketing	\$15.00	33	20	25	1	2010
2	Database	\$15.00	24	20	25	2	2011
3	Economics	\$20.00	24	20	25	2	2010
4	Accountancy	\$20.00	35	20	25	3	2014
5	Digital Marketing	\$25.00	25	20	25	3	2015
6	Big Data	\$25.00	20	20	25	4	2016
7	Management	\$32.00	22	20	25	5	2017
8	Ethics	\$35.00	19	20	25	5	2012
9	GDPR	\$40.00	20	20	25	4	2010
10	Maths	\$40.00	19	20	25	4	2009
11	Language	\$25.00	35	20	25	3	2009
12	Humanities	\$30.00	23	20	25	5	2008
13	AI	\$10.00	25	0	0	1	2011
14	Fishing	\$20.00	25	10	20	2	2009
(New)		\$0.00	0	0	0	0	0

Figure 38: The book's stock level before the update.

Actual:

BookID	BookName	CoverPrice	StockLevel	ReorderLev	TargetLeve	PublisherID	Publication
1	Marketing	\$15.00	33	20	25	1	2010
2	Database	\$15.00	24	20	25	2	2011
3	Economics	\$20.00	24	20	25	2	2010
4	Accountancy	\$20.00	35	20	25	3	2014
5	Digital Marketing	\$25.00	25	20	25	3	2015
6	Big Data	\$25.00	20	20	25	4	2016
7	Management	\$32.00	32	20	25	5	2017
8	Ethics	\$35.00	19	20	25	5	2012
9	GDPR	\$40.00	20	20	25	4	2010
10	Maths	\$40.00	19	20	25	4	2009
11	Language	\$25.00	35	20	25	3	2009
12	Humanities	\$30.00	23	20	25	5	2008
13	AI	\$10.00	25	0	0	1	2011
14	Fishing	\$20.00	25	10	20	2	2009
(New)		\$0.00	0	0	0	0	0

Figure 39: The output after updating stock level.

PurchaseOrderFromPubID	PurchaseDate	PublisherID	StaffID	PurOrderPubStatus	Click to Add
1	2023/12/8	1	1	☒	
15	2023/9/11	2	2	☒	
19	2023/9/21	2	2	☒	
20	2023/9/13	2	2	☒	
21	2023/12/4	3	2	☒	
22	2023/5/16	4	2	☒	
23	2023/9/13	4	2	☒	
24	2023/9/5	3	2	☒	
25	2023/11/27	3	2	☒	
26	2023/12/4	1	2	☒	
27	2023/12/5	5	2	☒	
(New)		0	0	☐	

Figure 40: The output after updating the purchase order status.

The final invoice also matches.

PurchaseOrderFrom	27
PurchaseDate	2023/12/5
PublisherName	Xinhua
BookName	Management
CoverPrice	\$32.00
ReorderQuantity	10
Cost	¥320.00
Discount	0
Total	32

Figure 41: The final discount invoice of new data.

ID2.8 Reflection

2.8.1 Advantages

Enhancements to inventory management via additional fields enable the system to prompt timely restocking, thus optimizing inventory efficiency. Furthermore, introducing order status tracking functionalities compresses the lead time from procurement to sales, enhancing customer satisfaction and response speed. Analyses of customer value data equip managerial staff with the tools to identify and maintain high-value customers, crucial for personalized service and bolstering customer loyalty. Additionally, leveraging purchase quantity data allows management to evaluate and refine supplier relationships, ensuring supply chain stability and potentially securing more favorable bulk purchase terms.

2.8.2 Disadvantages

Data Entry Challenges

Manual entry in Microsoft Access is time-consuming and error-prone, especially in

peak periods.

Security Concerns

The lack of a password recovery option in Access affects convenience and security.

2.8.3 Solutions and self-evaluation

Data Automation

Use Access to import pre-processed data from Excel, text files, or other databases.

Set data validation rules in table fields to minimize errors.

Enhanced Security

Develop custom password management and reset features using VBA programming in Access.

Integrate Access with external systems or applications for multi-factor authentication (MFA).

Self-evaluation

In familiarizing with the system, I've learned the importance of database operations and attention to detail, enhancing my data analysis and problem-solving skills.

References

Chiru, G. (2015) 'Logistical stock management methods - organizational performance evaluation means', *Journal of Accounting & Management* (2284-9459), 5(3), pp. 38–46. Available at:

<https://search-ebscohost-com-s.elink.xjtlu.edu.cn:443/login.aspx?direct=true&db=bsu&AN=112714793&site=eds-live&scope=site> (Accessed: 16 December 2023).

Jiang, L. and Naumann, F. (2020) 'Holistic primary key and foreign key detection', *Journal of Intelligent Information Systems: Integrating Artificial Intelligence and Database Technologies*, 54(3), pp. 439–461. doi:10.1007/s10844-019-00562-z.